

# Md Biplob Hosen

PHD STUDENT · INFORMATION SYSTEMS

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## Education

### University of Maryland Baltimore County (UMBC)

Baltimore, MD, USA

PH.D. IN INFORMATION SYSTEMS

Aug 2024 – Present

- Advisor: Dr. Lujie Karen Chen

### University of Maryland Baltimore County (UMBC)

Baltimore, MD, USA

M.S. IN INFORMATION SYSTEMS

Aug 2024 – 2026

- Advisor: Dr. Lujie Karen Chen
- CGPA: 4.00/4.00

### Jahangirnagar University

Savar, Dhaka, Bangladesh

M.S. IN INFORMATION TECHNOLOGY (IT)

2016 – 2017

- Thesis Group
- CGPA: 3.82/4.00

### Jahangirnagar University

Savar, Dhaka, Bangladesh

B.SC. (HONS.) IN INFORMATION TECHNOLOGY (IT)

2013 – 2016

- CGPA: 3.81/4.00

## Professional Experience

- 2024-Now **Graduate Assistant**, Information Systems, UMBC
- 2023-Now **Assistant Professor**, IIT, Jahangirnagar University
- 2021-2023 **Lecturer**, IIT, Jahangirnagar University
- 2020-2021 **Technical Executive**, Teletalk Bangladesh Limited
- 2018-2020 **Lecturer**, CSE, Stamford University Bangladesh
- 2019-2019 **Adjunct Faculty**, ICT, Bangladesh University of Professionals
- 2017-2020 **Adjunct Faculty**, CSE, Sonargoan University
- 2017 **Lecturer**, CSE, Daffodil International University
- 2017 **Guest Lecturer**, City University
- 2017 **Teacher Assistant**, IIT, Jahangirnagar University

## Publications

### RESEARCH ARTICLES

- Hosen, M. B.**, et al. 2026. HealthNLP\_Retrievers at ArchEHR-QA 2026: Cascaded LLM Pipeline for Grounded Clinical Question Answering. In Proceedings of the 3rd Workshop on Patient-Oriented Language Processing (CL4Health) at LREC 2026, May 11-16, Palma, Mallorca (Spain).
- Hosen, M. B.**, et al. 2026. NeuroSymRead: Symbolic Governance of Neural Generation for Adaptive Dialogic Reading. In Proceedings of the XAI-Ed 2026 Workshop: Demystifying AI in Education and Learning Analytics through Explainability, Agency, and Transparency, held at the 16th International Conference on Learning Analytics and Knowledge (LAK 2026), April 27–28, Bergen, Norway.
- Hosen, M. B.**, et al. 2026. SQU: Sensitivity-Guided Quantum Machine Unlearning for Privacy-Preserving ML. *IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*.
- Hosen, M. B.**, et al. 2026. Unified Bangla: A Transformer-Based Framework for Bangla Dialect Identification and Cross-Dialectal Translation. *IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*.

- Hosen, M. B.**, et al. 2025. Modeling Socially Constructed Knowledge using Multimodal Machine Learning: A Case Study in K-12 AI Literacy Education Classroom. *26th International Conference on Artificial Intelligence in Education (AIED)*.
- Hosen, M. B.**, et al. 2025. Assessing and Mitigating Intersectional Bias: A Multi-Domain Study on Race and Gender. *IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS)*.
- Hosen, M. B.**, et al. 2025. A Hybrid Approach towards Multi-Label Semi-Supervised Emotion Classification. *IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS)*.
- Hosen, M. B.**, et al. 2025. Explainable AI Based Multi-Class Pneumonia Classification and Anomaly Detection from Chest X-ray. *IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS)* (Accepted).
- Hosen, M. B.**, et al. 2025. From Code to Career: Assessing Competitive Programmers for Industry Placement. *2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*.
- Hosen, M. B.**, et al. 2025. Image captioning in Bengali language using visual attention. *PLOS ONE*, 20(2).
- Hosen, M. B.**, et al. 2024. Bias in Deep Learning Skin Cancer Detection. *Proceedings of the 3rd International Conference on Computing Advancements*, 993-1000.
- Hosen, M. B.**, et al. 2024. Impact, Causation and Prediction of Socio-Academic and Economic Factors in Exam-centric Student Evaluation. *13th International Conference on Electrical and Computer Engineering (ICECE)*, 321-326. IEEE.
- Hosen, M. B.**, et al. 2024. Effects of Movement Restrictions on Consumer Consumption. *Journal of Applied Engineering and Technological Science (JAETS)*, 6(1): 416-428.
- Hosen, M. B.**, et al. 2023. Eligible Personal Loan Applicant Selection using Federated Machine Learning Algorithm. *International Journal of Advanced Computer Science and Applications*, 14(8).
- Hosen, M. B.**, et al. 2022. Quality analysis of pathao ride-sharing service in Bangladesh. *International Joint Conference on Advances in Computational Intelligence*, 597-611. Springer.
- Hosen, M. B.**, 2022. Impact of Telecommunication Service Quality in Bangladesh on Online Education during Covid-19. *International Journal of Modern Education and Computer Science*, 10(6), 65, 597-611. Springer.
- Hosen, M. B.**, 2022. Performance Analysis of Machine Learning Techniques in Network Intrusion Detection. *Journal of Information Technology (JUJIT)*, 11.
- Hosen, M. B.**, et al. 2022. Secondary User Channel Selection in Cognitive Radio Network Using Adaptive Method. *2018 3rd International Conference for Convergence in Technology (I2CT)*, Pune, India, 2018, pp. 1-6.

## Awards, Fellowships, & Grants

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- 2023-2024    **Research Grant**, Jahangirnagar University
- 2022-2023    **Research Grant**, Jahangirnagar University
- 2021-2022    **Research Grant**, Jahangirnagar University
- 2018        **National Science & Technology (N.S.T.) Fellowship**, Ministry of Science and Technology, BD
- 2013-2016    **Regular Merit-Based Annual Scholarships**, Jahangirnagar University  
**Education Board Scholarship**, Higher Secondary Certificate (HSC) Result  
**Education Board Scholarship**, Secondary School Certificate (SSC) Result

## Mentoring

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- 2023-2024    **Md Imranur Rahman Akib, Fathima Binthe Muhammed, Umit Saha**, B.Sc. Student, Jahangirnagar University
- 2022-2023    **Faria Ahmed**, M.S. Student, Jahangirnagar University
- 2021-2022    **Md. Uthba Bin Anowar Binoy**, M.S. Student, Jahangirnagar University
- 2020-2021    **Md. Habibullah**, M.S. Student, Jahangirnagar University

## Service & Professional Development

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### UNIVERSITY SERVICE & LEADERSHIP

- 2021-2024 **Student Residential Hall, Jahangirnagar University**, House Tutor
- 2021-2024 **Digital Filing System Implementation Committee, Jahangirnagar University**, Member
- 2019-2020 **Stamford University Bangladesh**, Student Advisor
- 2016-2017 **Jahangirnagar University Bogra Student Association**, Vice-President
- 2016-2017 **Jahangirnagar University IT Society (JUITs)**, Joint Secretary

2016-2017

### DEVELOPMENT

- 2021 **Teacher's Induction Program**, Organized by IQAC, Jahangirnagar University. Training focused on tools and methods for quality improvement in higher education.
- 2018 **Faculty Training Program**, Organized by IQAC, Stamford University Bangladesh. Professional development workshop for university faculty members.
- 2016 **Advanced Mobile Game Development.**, Technical training provided by the ICT Division, Ministry of Posts, Telecommunications and Information Technology, Bangladesh.
- 2015 **Software Testing & Automation**, Workshop held at the Institute of Information Technology (IIT), Jahangirnagar University.

### REVIEWER

- 2026 **27th International Conference on Artificial Intelligence in Education (AIED2026)**,
- 2025 **IEEE Open Journal of the Computer Society**,
- 2025 **2026 Learning Engineering Research Network (LERN) Convening**,
- 2025 **3rd World Congress on Smart Computing (WCSC 2026)**,
- 2025 **IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS 2025)**,
- 2024 **International Conference on Computing and Machine Learning (CML 2024)**,
- 2022 **IEEE World Conference on Applied Intelligence and Computing (AIC 2022)**,
- 2022 **6th Intl. Joint Conference on Advances in Computational Intelligence (IJCACI 2022)**,

### MEMBERSHIP

- 2026 **Phi Kappa Phi, UMBC Chapter** , Member
- 2016 **IEEE Student Branch, Jahangirnagar University**, Volunteer
- 2016 **Jahangirnagar University Science Club** , Volunteer